INTERNET PROGRAMMING LAB

Experiment No. 11

Title: Online Real Estate Portal (Home Finder)

AIM

To design and develop an online real estate web application using HTML, CSS and JavaScript that lets buyers register and login, search properties by location, type, price, bedrooms and amenities, save and reuse search criteria, view results on a map, schedule viewings, save favourites, search agents/companies and calculate mortgage EMI.

ALGORITHM

Step 1. Start and display the Buyer Login / Register page with advertisement banner; increment and store the website visitor count.

Step 2. Register: validate and store username and password; Login: verify credentials.

Step 3. Search: read location, property type, maximum price, minimum bedrooms and selected amenities.

Step 4. Filter the property list so every condition is satisfied and display title, location, price, amenities and agent.

Step 5. Map View: draw a pin for every result; clicking a pin highlights the corresponding listing (Google Maps link is provided for each listing).

Step 6. Save Search: store the current criteria in the logged-in buyer's record; Use Saved Search: load the stored criteria into the form and search again.

Step 7. Favourites: buyer can save or remove a property from favourites.

Step 8. Schedule Viewing: store the property, date and time and notify the agent.

Step 9. Agent & Company Search: filter the agent table by name, company or area.

Step 10. Mortgage Calculator: read loan amount, rate and years; compute EMI = P x r x (1+r)^n / ((1+r)^n - 1) where r = rate/12/100 and n = months.

Step 11. My Saved page lists favourites and scheduled viewings; logout and stop.

PROGRAM

File: p11.html (HTML structure, CSS styling and JavaScript logic in a single file)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>Home Finder - Property Search</title>
<style>
:root{--c:#b4531a}
*{box-sizing:border-box;margin:0;padding:0;font-family:Segoe UI,Arial,sans-serif}
body{background:#f2f5fa;color:#222;line-height:1.4}
header{background:var(--c);color:#fff;padding:12px 20px;display:flex;justify-content:space-between;align-
items:center;flex-wrap:wrap;gap:8px}
nav button{background:none;border:0;color:#fff;margin-left:4px;padding:7px 11px;cursor:pointer;border-
radius:4px;font-size:14px}
nav button:hover{background:#ffffff33}
main{max-width:1000px;margin:18px auto;padding:0 12px}
.card{background:#fff;border-radius:8px;padding:16px;margin-bottom:16px;box-shadow:0 1px 4px #0002}
.page{display:none}.page.on{display:block}
h3{margin-bottom:10px;color:var(--c)}
input,select,textarea{padding:7px;margin:4px 6px 4px 0;border:1px solid #bbb;border-radius:4px;font-
size:14px}
.btn{background:var(--c);color:#fff;border:0;padding:8px 14px;border-radius:4px;cursor:pointer;font-
size:14px;margin:2px}
.btn.red{background:#b3261e}.btn.gray{background:#666}
table{width:100%;border-collapse:collapse;margin-top:8px;font-size:14px}
th,td{border:1px solid #ddd;padding:7px;text-align:left}th{background:#e9eef6}
.ok{color:#0a7a2f;margin:6px 0}.err{color:#b3261e;margin:6px 0}
.grid{display:grid;grid-template-columns:repeat(auto-fill,minmax(215px,1fr));gap:12px}
.item{border:1px solid #e1e4ea;border-radius:8px;padding:12px;background:#fff}
.pic{height:90px;border-radius:6px;display:flex;align-items:center;justify-content:center;font-
size:44px;background:linear-gradient(135deg,#dfe9f7,#f7e9df);margin-bottom:8px}
.pill{display:inline-block;background:#e5ecf8;border-radius:12px;padding:2px 10px;font-size:12px;margin:2px}
.stat{display:inline-block;min-width:150px;background:#eef3fb;border-radius:8px;padding:10px
14px;margin:4px;text-align:center}
.stat b{display:block;font-size:22px;color:var(--c)}
@media(max-width:600px){nav button{padding:5px 6px;font-size:12px}table{font-size:12px}}

.ad{background:#fff3cd;border:1px solid #e6c55b;text-align:center;font-size:13px;padding:6px;border-
radius:6px;margin-bottom:12px}
.pin{cursor:pointer}.pin circle{fill:#b4531a;stroke:#fff;stroke-width:2}.pin.hl circle{fill:#1a73e8;r:12}
.two{display:grid;grid-template-columns:1fr 1fr;gap:14px}
@media(max-width:700px){.two{grid-template-columns:1fr}}
</style>
</head>
<body>
<header>
  <h2>&#127968; Home Finder</h2>
  <nav>
    <button onclick="go('search')">Find Property</button>
    <button onclick="go('agents')">Agents &amp; Companies</button>
    <button onclick="go('tools')">Mortgage Calculator</button>
    <button onclick="go('mine')">My Saved</button>
    <button id="authBtn" onclick="authClick()">Buyer Login</button>
  </nav>
</header>
<main>
  <div class="ad">Advertisement: <a href="#" onclick="return false">SafeHome Insurance &mdash; 20% off first
year</a>  |  <a href="#" onclick="return false">QuickLoan Bank &mdash; 8.1% home loans</a></div>

  <section id="login" class="page card">
    <h3>Buyer Login / Register</h3>
    <input id="lu" placeholder="Username"><input id="lp" type="password" placeholder="Password">
```

```
    <button class="btn" onclick="login()">Login</button><button class="btn gray"
onclick="reg()">Register</button>
    <p id="lmsg"></p>
  </section>

  <section id="search" class="page">
    <div class="card">
      <h3>Search Properties</h3>
      Location <select id="floc"></select>
      Type <select id="ftype"><option>Any</option><option>House</option><option>Apartment</
option><option>Land</option><option>Commercial</option></select>
      Max price (Lakhs) <input type="number" id="fmax" style="width:90px" placeholder="e.g. 80">
      Min bedrooms <select id="fbed"><option
value="0">Any</option><option>1</option><option>2</option><option>3</option><option>4</option></select><br>
      Amenities:
      <label><input type="checkbox" class="am" value="Parking"> Parking</label>
      <label><input type="checkbox" class="am" value="Garden"> Garden</label>
      <label><input type="checkbox" class="am" value="Gym"> Gym</label>
      <label><input type="checkbox" class="am" value="Lift"> Lift</label><br>
      <button class="btn" onclick="doSearch()">Search</button>
      <button class="btn gray" onclick="saveCriteria()">Save this search</button>
      <button class="btn gray" onclick="useSaved()">Use saved search</button>
      <span id="smsg" class="ok"></span>
    </div>
    <div class="two">
      <div class="card"><h3 id="rtitle">Results</h3><div id="plist"></div></div>
      <div class="card"><h3>Map View</h3><svg id="map" viewBox="0 0 400 320" width="100%"
style="background:#e8f1e4;border-radius:8px"></svg>
        <small>Click a pin to locate the listing. (Schematic map; link to Google Maps available on each
listing.)</small></div>
    </div>
  </section>

  <section id="agents" class="page card">
    <h3>Agent &amp;amp; Company Search</h3>
    <input id="aq" placeholder="Agent / company / area" oninput="renderAgents()" style="width:260px">
    <div id="alist"></div>
  </section>

  <section id="tools" class="page card">
    <h3>Mortgage (EMI) Calculator</h3>
    Loan amount (Rs.) <input type="number" id="la" value="4000000">
    Interest % p.a. <input type="number" id="lr" value="8.5" style="width:70px">
    Years <input type="number" id="ly" value="20" style="width:60px">
    <button class="btn" onclick="emi()">Calculate</button>
    <p id="emiout" style="margin-top:8px"></p>
  </section>

  <section id="mine" class="page card">
    <h3>My Favourites &amp;amp; Scheduled Viewings</h3><div id="mlist"></div>
  </section>
</main>

<script>
const $ = id => document.getElementById(id);
const load = (k, d) => { try { return JSON.parse(localStorage.getItem(k)) || d; } catch (e) { return d; } };
const save = (k, v) => { try { localStorage.setItem(k, JSON.stringify(v)); } catch (e) {} };

const props = [
  {id:1, title:'3BHK Villa with Garden',  loc:'Anna Nagar',  type:'House',      price:95, bed:3, am:
['Parking','Garden'],        agent:'Ravi Menon', x:90,  y:80,  icon:'&#127969;'},
  {id:2, title:'2BHK Apartment',          loc:'Velachery',   type:'Apartment',  price:58, bed:2, am:
['Parking','Lift','Gym'],    agent:'Sana Ali',   x:250, y:110, icon:'&#127970;'},
  {id:3, title:'1BHK Studio Flat',        loc:'Tambaram',    type:'Apartment',  price:32, bed:1, am:
['Lift'],                    agent:'Sana Ali',   x:310, y:230, icon:'&#127970;'},
  {id:4, title:'1200 sq.ft Residential Plot', loc:'Anna Nagar', type:'Land',     price:70, bed:0, am:[],
agent:'Ravi Menon', x:140, y:170, icon:'&#127795;'},
  {id:5, title:'4BHK Duplex House',       loc:'Velachery',   type:'House',      price:140,bed:4, am:
['Parking','Garden','Gym'],  agent:'Kavita Rao', x:200, y:260, icon:'&#127968;'},
```

```
  {id:6, title:'Shop Space - Main Road',  loc:'Tambaram',    type:'Commercial', price:85, bed:0, am:
['Parking'],                 agent:'Kavita Rao', x:60,  y:250, icon:'&#127978;'}
];
const agents = [
  {n:'Ravi Menon', co:'PrimeNest Realty',  area:'Anna Nagar', ph:'98400 11111'},
  {n:'Sana Ali',   co:'UrbanKey Estates',  area:'Velachery, Tambaram', ph:'98400 22222'},
  {n:'Kavita Rao', co:'PrimeNest Realty',  area:'Velachery, Tambaram', ph:'98400 33333'}
];
let users = load('hf_users', []), favs = load('hf_favs', []), views = load('hf_views', []), me = null, shown
= [], hl = 0;
let visits = load('hf_visits', 0) + 1; save('hf_visits', visits);

function msg(id, t, bad) { $(id).className = bad ? 'err' : 'ok'; $(id).textContent = t; }
function go(p) {
  if (p === 'mine' && !me) { msg('lmsg', 'Please login first.', 1); p = 'login'; }
  document.querySelectorAll('.page').forEach(s => s.classList.remove('on')); $(p).classList.add('on');
  if (p === 'mine') renderMine(); if (p === 'agents') renderAgents();
}
function reg() {
  const u = $('lu').value.trim(), p = $('lp').value;
  if (!u || p.length < 4) return msg('lmsg', 'Username and 4+ char password required.', 1);
  if (users.some(x => x.u === u)) return msg('lmsg', 'User exists.', 1);
  users.push({u, p, crit:null}); save('hf_users', users); msg('lmsg', 'Registered. Please login.');
}
function login() {
  const f = users.find(x => x.u === $('lu').value.trim() && x.p === $('lp').value);
  if (!f) return msg('lmsg', 'Invalid credentials.', 1);
  me = f; $('authBtn').textContent = 'Logout (' + f.u + ')'; go('search');
}
function authClick() { if (me) { me = null; $('authBtn').textContent = 'Buyer Login'; } go('login'); }

/* ---------- search ---------- */
function init() {
  $('floc').innerHTML = '<option>Any</option>' + [...new Set(props.map(p => p.loc))].map(l => `<option>$
{l}</option>`).join('');
}
function criteria() {
  return {loc:$('floc').value, type:$('ftype').value, max:$('fmax').value, bed:$('fbed').value, am:
[...document.querySelectorAll('.am:checked')].map(c => c.value)};
}
function doSearch() {
  const c = criteria();
  shown = props.filter(p => (c.loc === 'Any' || p.loc === c.loc) && (c.type === 'Any' || p.type === c.type)
&&
    (!c.max || p.price <= +c.max) && p.bed >= +c.bed && c.am.every(a => p.am.includes(a)));
  $('rtitle').textContent = `Results (${shown.length})`;
  $('plist').innerHTML = shown.map(p => `<div class="item" style="margin-bottom:8px;${p.id === hl ?
'border:2px solid #1a73e8' : ''}">
    <div style="display:flex;gap:10px"><div class="pic" style="width:80px;height:70px;font-
size:34px;margin:0">${p.icon}</div>
    <div><b>${p.title}</b><br>${p.loc} &bull; ${p.type}${p.bed ? ' &bull; ' + p.bed + ' BHK' : ''}<br><span
class="price" style="color:#b4531a;font-weight:bold">Rs. ${p.price} Lakhs</span><br>
    ${p.am.map(a => `<span class="pill">${a}</span>`).join('')}</div></div>
    <small>Agent: ${p.agent}</small><br>
    <button class="btn" onclick="fav(${p.id})">${favs.includes(p.id) ? '&#9829; Saved' : '&#9825;
Save'}</button>
    <button class="btn gray" onclick="viewing(${p.id})">Schedule viewing</button>
    <a href="https://www.google.com/maps/search/?api=1&query=${encodeURIComponent(p.loc + ' Chennai')}"
target="_blank">Google Maps</a></div>`).join('') || '<p class="err">No properties match your criteria.</p>';
  drawMap();
}
function drawMap() {
  let s = '<path d="M0 150 Q100 120 200 160 T400 140" stroke="#9cc3e6" stroke-width="10" fill="none"/><path
d="M200 0 L200 320" stroke="#fff" stroke-width="6"/><path d="M0 220 L400 200" stroke="#fff" stroke-
width="6"/>';
  shown.forEach(p => s += `<g class="pin ${p.id === hl ? 'hl' : ''}" onclick="hl=${p.id};doSearch()"><circle
cx="${p.x}" cy="${p.y}" r="9"/><text x="${p.x + 12}" y="${p.y + 4}" font-size="11"
fill="#333">${p.loc}</text></g>`);
  $('map').innerHTML = s;
```

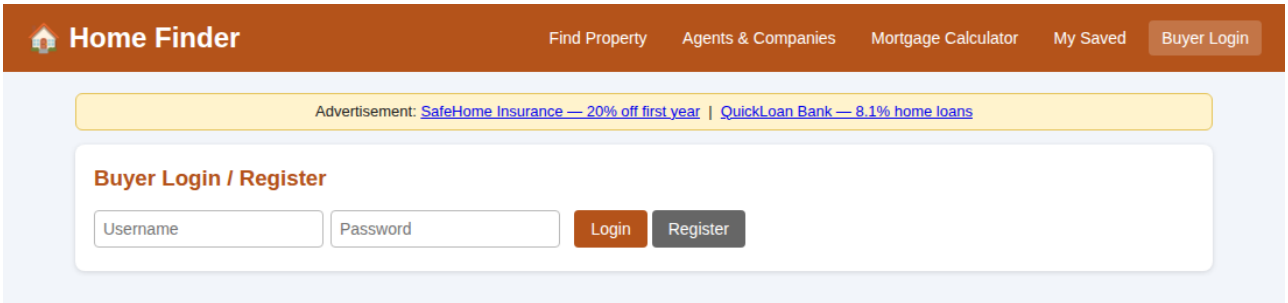
```
}
function saveCriteria() {
  if (!me) { msg('lmsg', 'Login to save your search criteria.', 1); return go('login'); }
  me.crit = criteria(); save('hf_users', users); msg('smsg', ' Search criteria saved.');
}
function useSaved() {
  if (!me || !me.crit) return msg('smsg', ' No saved criteria found (login and save one first).', 1);
  const c = me.crit; $('floc').value = c.loc; $('ftype').value = c.type; $('fmax').value = c.max; $
('fbed').value = c.bed;
  document.querySelectorAll('.am').forEach(a => a.checked = c.am.includes(a.value)); doSearch(); msg('smsg',
' Saved criteria applied.');
}

/* ---------- favourites / viewings ---------- */
function fav(id) {
  if (!me) { msg('lmsg', 'Login to save favourites.', 1); return go('login'); }
  favs = favs.includes(id) ? favs.filter(x => x !== id) : [...favs, id]; save('hf_favs', favs); doSearch();
}
function viewing(id) {
  if (!me) { msg('lmsg', 'Login to schedule a viewing.', 1); return go('login'); }
  const d = new Date(); d.setDate(d.getDate() + 2);
  views.push({pid:id, user:me.u, date:d.toISOString().slice(0, 10), time:'11:00'}); save('hf_views', views);
  msg('smsg', ' Viewing scheduled for ' + d.toISOString().slice(0, 10) + ' at 11:00. Agent notified.');
}
function renderMine() {
  const t = id => props.find(p => p.id === id).title;
  $('mlist').innerHTML = '<b>Favourites</b>' + (favs.length ? '<ul style="margin-left:20px">' + favs.map(i
=> `<li>${t(i)}</li>`).join('') + '</ul>' : '<p>None yet.</p>') +
    '<b>Scheduled viewings</b>' + (views.filter(v => v.user === me.u).length ?
'<table><tr><th>Property</th><th>Date</th><th>Time</th></tr>' +
    views.filter(v => v.user === me.u).map(v =>
`<tr><td>${t(v.pid)}</td><td>${v.date}</td><td>${v.time}</td></tr>`).join('') + '</table>' : '<p>None
yet.</p>');
}

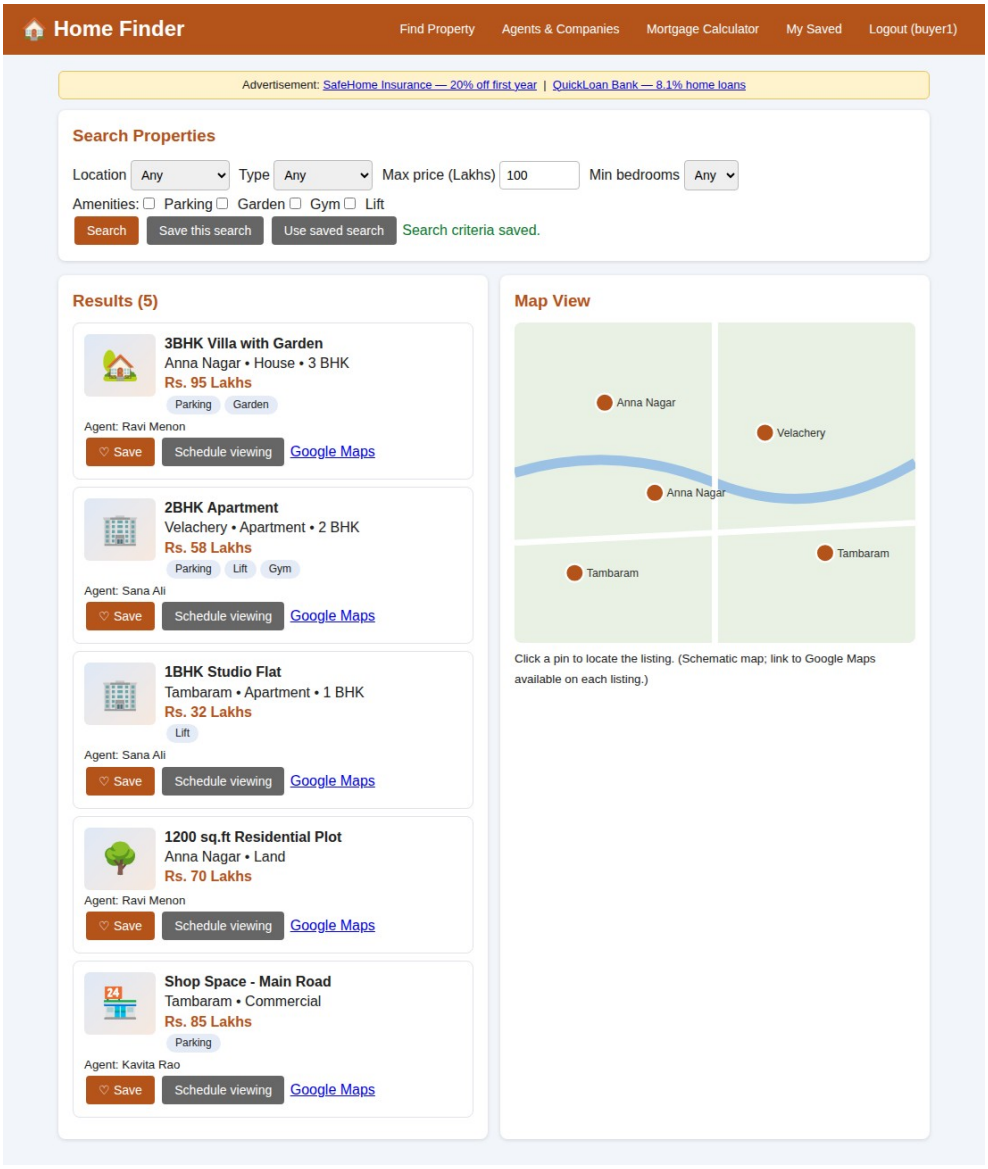
/* ---------- agents & EMI ---------- */
function renderAgents() {
  const q = $('aq').value.toLowerCase();
  const l = agents.filter(a => (a.n + a.co + a.area).toLowerCase().includes(q));
  $('alist').innerHTML = '<table><tr><th>Agent</th><th>Company</th><th>Areas</th><th>Phone</
th><th>Listings</th></tr>' +
    l.map(a => `<tr><td>${a.n}</td><td>${a.co}</td><td>${a.area}</td><td>${a.ph}</td><td>${props.filter(p =>
p.agent === a.n).length}</td></tr>`).join('') + '</table>';
}
function emi() {
  const P = +$('la').value, r = +$('lr').value / 1200, n = +$('ly').value * 12;
  if (!(P > 0 && r > 0 && n > 0)) return msg('emiout', 'Enter valid loan details.', 1);
  const e = P * r * Math.pow(1 + r, n) / (Math.pow(1 + r, n) - 1);
  $('emiout').className = 'ok';
  $('emiout').innerHTML = `Monthly EMI: <b>Rs. ${Math.round(e).toLocaleString('en-IN')}</b><br>Total
payable: Rs. ${Math.round(e * n).toLocaleString('en-IN')} | Total interest: Rs. ${Math.round(e * n -
P).toLocaleString('en-IN')}`;
}

init(); doSearch(); go('search');
$('authBtn').title = 'Website visits recorded: ' + visits;
</script>
</body>
</html>
```

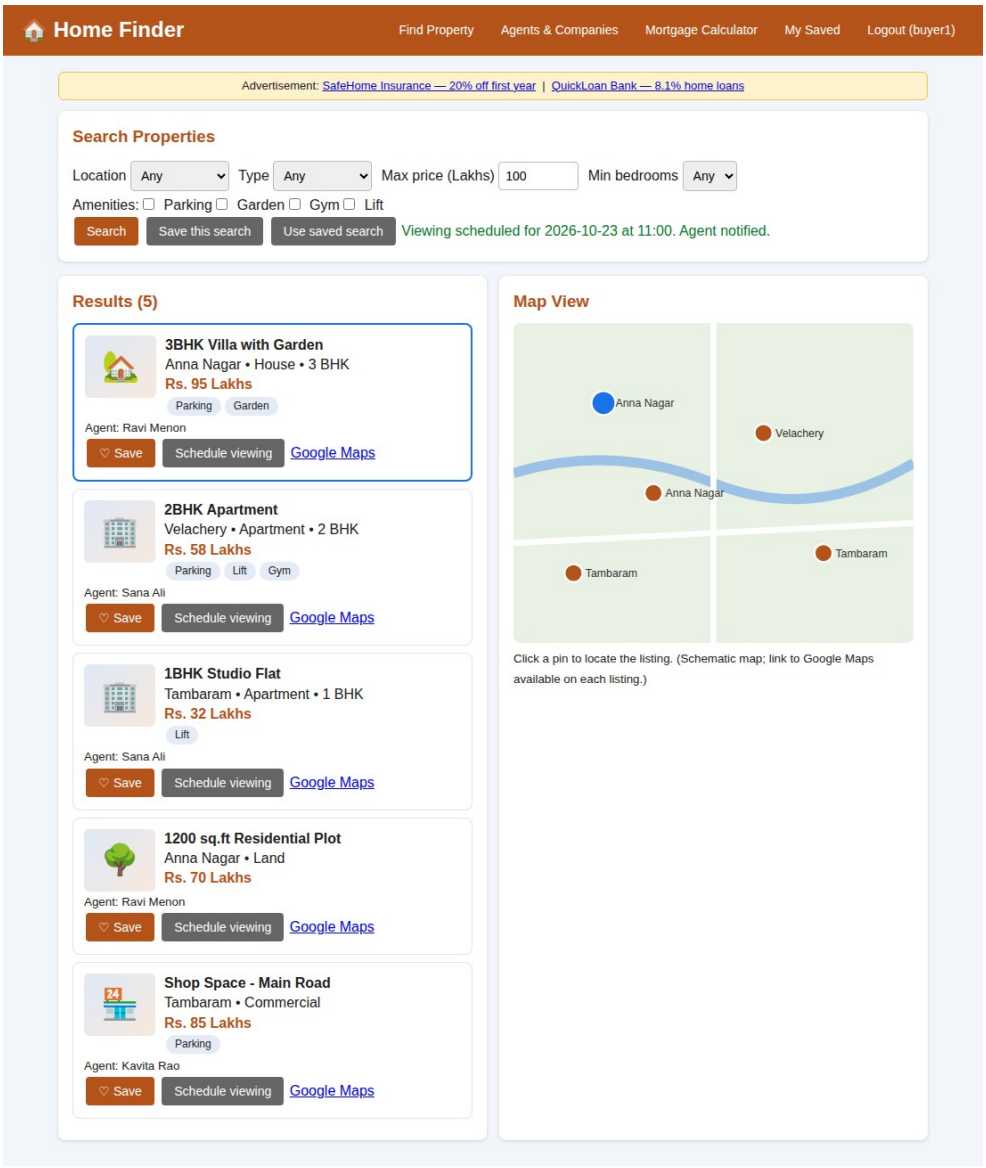
OUTPUT



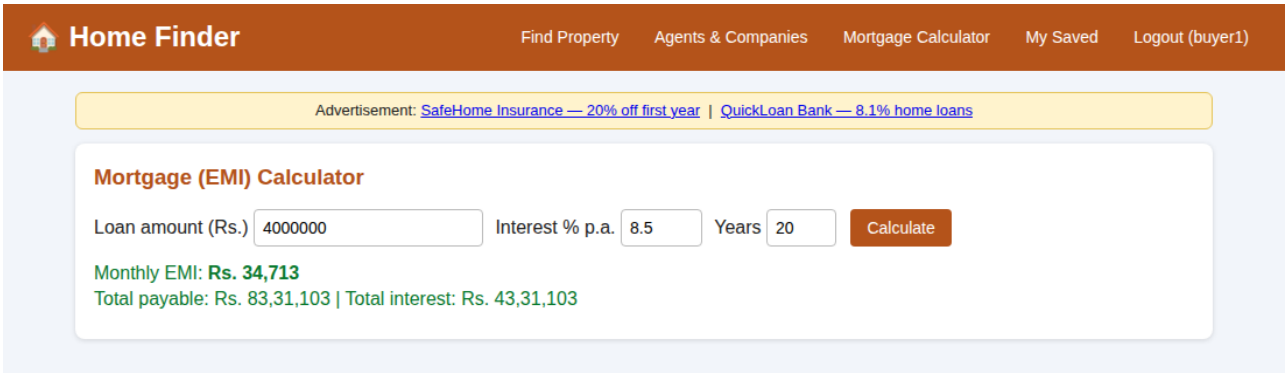
Buyer Login / Register page with advertisement



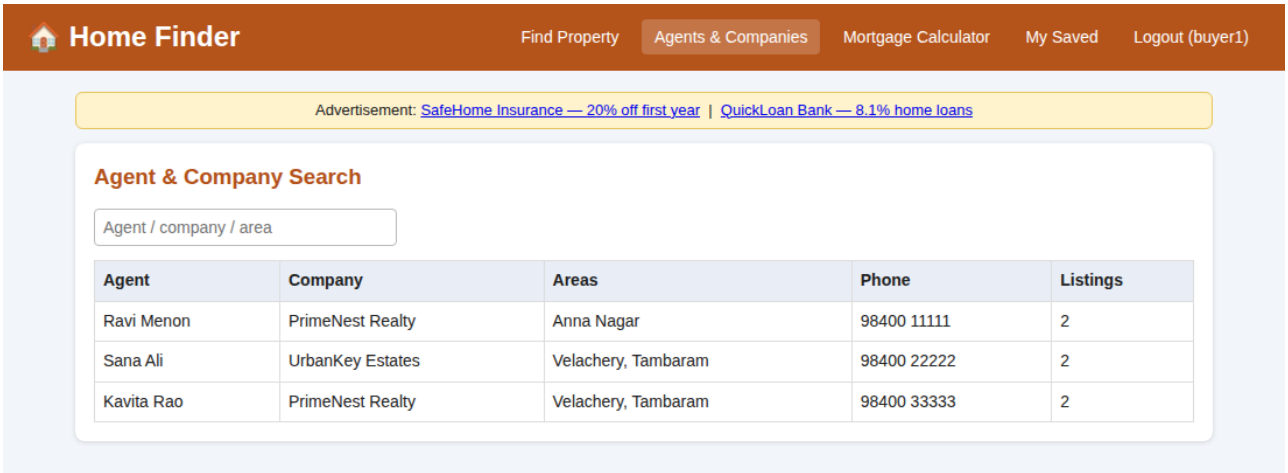
Property search results with map view and saved search criteria



Viewing scheduled and property located on map



Mortgage (EMI) calculator output



Agent and company search

RESULT

Thus the Home Finder real estate website was designed and implemented using HTML, CSS and JavaScript. Properties were searched with saved criteria, located on the map, viewings were scheduled and the EMI was calculated. The output was verified successfully.